1 Gain law (slope only)

(bars denote normalized quantities: lengths by R; $a_o := \Delta t/(\gamma_N R)$ [1/pN]; $\bar{a} := a/a_{\text{nom}}$; $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $\bar{w}$ = particle centre along the wall normal; $\bar{w}_s$ = the visible surface on the same axis, given (plane: the wall, contact at $h = 1$; cell: the visible top); $h := \bar{w} - \bar{w}_s$; p = fixed exponent; c = the one constant of the wall; the law is used on $h \geq 1.1$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}'(h;\, c) := \frac{1-c}{h^p}$$

$$\frac{\partial \bar{a}'}{\partial c} = -\frac{1}{h^p} = -\frac{\bar{a}'}{1-c}$$

$$\frac{\partial \bar{a}'}{\partial h} = -\frac{p\,\bar{a}'}{h}$$

$$\frac{\partial \bar{a}'}{\partial \bar{a}_w} = 0$$

$$\int_h^\infty \bar{a}'(s;\, c)\, ds = \frac{1-c}{(p-1)\,h^{p-1}}, \qquad p = 2,\ c = 0: \quad \int_1^\infty \bar{a}'\, ds = 1 \quad (\bar{a}(1) = 0,\ \bar{a}(\infty) = 1)$$

(no value law is written: $\bar{a}_w$ is a state; the law only gives its increment per step of height)

2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k-d+j-1],\ j = 1, \ldots, d+1,\ d = 2;\ \bar{a}'_{\text{true}}$ = the plant's slope, not in the family$\big)$

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}\big(\bar{w}[k] - \bar{w}_s\big)\,\Delta\bar{w}[k] + O(\Delta\bar{w}^2) \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
c[k+1] &= c[k]
\end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i]$$

$$\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] - (1-\lambda_c)\,n_w[k]$$

$$r'[k] := \bar{a}'_{\text{true}}\big(\bar{w}[k] - \bar{w}_s\big) - \bar{a}'\big(\bar{w}[k] - \bar{w}_s;\, c\big)$$

(slope residual of the family on this wall; zero only if the wall is in the family)

3 State vector

$\mathbf{x} = \big[\, \delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ c \,\big]^\top$ (slots 1–3 tracking error, 4 = the a slot, 5 = c)

4 Estimator

$\big(M[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$ = the increment the estimator believes the particle makes$\big)$

$$
\begin{aligned}
\hat{\bar{w}}[k] &= \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] \\
\hat{h}[k] &= \hat{\bar{w}}[k] - \bar{w}_s \\
\bar{a}'[k] &:= \bar{a}'\big(\hat{h}[k];\, \hat{c}[k]\big) = \frac{1-\hat{c}[k]}{\hat{h}[k]^p} \\
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \bar{a}'[k]\,M[k] \\
\hat{c}[k+1] &= \hat{c}[k]
\end{aligned}
$$

5 Error dynamics

$\big(\mathbf{e} = [\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_c \,]^\top$; first order, coefficients at the estimate; $\Delta\bar{w}[k] - M[k] = (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]$; $\bar{w}[k] - \hat{\bar{w}}[k] = -\,e_{\delta\bar{w}_3}[k]\big)$

$$
\bar{a}'_{\text{true}}(\bar{w}[k] - \bar{w}_s) - \bar{a}'[k] = \frac{p\,\bar{a}'[k]}{\hat{h}[k]}\,e_{\delta\bar{w}_3}[k] - \frac{1}{\hat{h}[k]^p}\,e_c[k] + r'[k]
$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \Big(\bar{a}'[k]\,(1-\lambda_c) + \frac{p\,\bar{a}'[k]}{\hat{h}[k]}\,M[k]\Big)\,e_{\delta\bar{w}_3}[k] + \big(1 + \bar{a}'[k]\,F_{dw}[k]\big)\,e_{\bar{a}_w}[k] \\
&\qquad - \frac{M[k]}{\hat{h}[k]^p}\,e_c[k] + \bar{a}'[k]\,\varepsilon_w[k] + M[k]\,r'[k] \\
e_c[k+1] &= e_c[k]
\end{aligned}
$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw}[k] & 0 \\ 0 & 0 & \bar{a}'[k]\,(1-\lambda_c) + \dfrac{p\,\bar{a}'[k]}{\hat{h}[k]}\,M[k] & 1 + \bar{a}'[k]\,F_{dw}[k] & -\dfrac{M[k]}{\hat{h}[k]^p} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$F_{e,44}[k] = 1 + \bar{a}'[k]\,F_{dw}[k]$ (no $\frac{\bar{a}''}{\bar{a}'}M$ term: the slope is read at $\hat{h}$, not at $\hat{\bar{a}}_w$)

$F_{e,45}[k] = -\dfrac{M[k]}{\hat{h}[k]^p}$ (the only place c acts; proportional to the step)

$M[k]\,r'[k]$ (the only entrance of the model error; zero in a hold)

6 Measurement

$\big(\nabla_d \bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]$; the reading is the gain d steps ago, walked back from the estimator height$\big)$

$y_1[k] = \delta \bar{w}_1[k] + n_w[k]$

$y_2[k] = \bar{a}_w[k-d] + n_a[k]$

$\hat{y}_1[k] = \delta \hat{\bar{w}}_1[k]$

$\hat{y}_2[k] = \hat{\bar{a}}_w[k] - \bar{a}'[k]\,\nabla_d \bar{w}_d[k]$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{p\,\bar{a}'[k]}{\hat{h}[k]}\,\nabla_d \bar{w}_d[k] & 1 & +\dfrac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^p} \end{bmatrix}$$

$H_{24}[k] = 1$ (y_2 reads the state $\bar{a}_w$ directly; the level is measured, not computed)

$H_{25}[k] = \dfrac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^p}$ (walk-back only; zero in a hold; may be set to 0 to keep c out of y_2 entirely)

$$\mathbf{e}_y[k] = H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

7 Noise

$Q[k] = Q_{33}[k]\,\mathbf{g}\,\mathbf{g}^\top, \qquad \mathbf{g} = \big[\,0,\ 0,\ -1,\ \bar{a}'[k],\ 0\,\big]^\top, \qquad Q_{55} = 0$

$Q_{33}[k] = \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \dfrac{4k_B T\,a_o}{R}$

$R[k] = \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \dfrac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k] + \xi\big)^2$

8 Observability

($\Phi(k,k_0) := F_e[k-1]\cdots F_e[k_0]$, $\Phi(k_0,k_0)=I$; N window length)

Gate 1, the two columns of e_c:

$$F_{e,45}[k] = -\frac{M[k]}{\hat{h}[k]^p} \quad \text{(state path, through } e_{\bar{a}_w} \text{ then } y_1, y_2)$$

$$H_{25}[k] = \frac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^p} \quad \text{(measurement path, walk-back only)}$$

Gate 2, time signatures of the pair ($e_{\bar{a}_w},\ e_c$):

$$e_c[k+1] = e_c[k] \qquad \text{(constant)}$$

$$e_{\bar{a}_w}[k+1] = \big(1+\bar{a}'[k]F_{dw}[k]\big)\, e_{\bar{a}_w}[k] - \frac{M[k]}{\hat{h}[k]^p}\, e_c[k] + \dots$$

(a level, drifting in proportion to the height travelled when $e_c \neq 0$)

$$y_2[k] = e_{\bar{a}_w}[k-d] + \dots \qquad \text{(the level is read every step, in a hold as well)}$$

Gate 3, $\det O_2$ of the pair with coefficients frozen ($\alpha := 1+\bar{a}'F_{dw}$, $\beta := M/\hat{h}^p$, $\gamma := \nabla_d\bar{w}_d/\hat{h}^p$, $q := -F_{dw}$):

$$A_2 = \begin{bmatrix} \alpha & -\beta \\ 0 & 1 \end{bmatrix}$$

$$\text{row } y_2\text{:}\quad C_2 = \begin{bmatrix} 1,\ \gamma \end{bmatrix}$$

$$\det \begin{bmatrix} C_2 \\ C_2A_2 \end{bmatrix} = -\beta - \gamma(\alpha-1) = -\frac{M + \nabla_d\bar{w}_d\,\bar{a}'F_{dw}}{\hat{h}^p}$$

row y_1 (through $F_{e,34} = -F_{dw}$): $\quad C_2 = \begin{bmatrix} q,\ 0 \end{bmatrix}$

$$\det \begin{bmatrix} C_2 \\ C_2A_2 \end{bmatrix} = -q^2\beta = -\frac{F_{dw}^2\, M}{\hat{h}^p}$$

$\det O_2 = 0$ (y_2) $\iff$ $M=0$ and $\nabla_d\bar{w}_d = 0$ (hold) or $\hat{h}\to\infty$

$\det O_2 = 0$ (y_1) $\iff$ $F_{dw}=0$ or $M=0$ (hold) or $\hat{h}\to\infty$

no interior zero: $\dfrac{\partial\bar{a}'}{\partial c} = -\hat{h}^{-p} \neq 0$ on the band (contrast formB, $J_b = 0$ at $\bar{w} = 1+\hat{b}$)

no alias with $\bar{a}_w$: $e_{\bar{a}_w}$ is read directly ($H_{24}=1$), e_c only through the step M

Seed and prior

fit of the slope on the band $h \in [1.10,\ 22.2]$:

plane, $c = 0 \ \Rightarrow \ p = 2.042$

(sub-segments $[1.1, 2] : 2.166,\ [2, 5.5] : 1.982,\ [5.5, 22.2] : 1.953$)

p fixed: $c_{\text{plane}} = +0.037, \quad c_{\text{cell}} = +0.636, \quad c_{\text{sphere}} = -0.062$

slope residual left by the best constant, $\sup|r'|$ on the band:

plane 0.024, cell 0.087, sphere 0.011

the constant the wall asks for along the band, $c_{\text{eff}}(h) = 1 - h^p \bar{a}'_{\text{true}}(h)$:

plane $+0.06 \to -0.28$, cell $+0.74 \to -0.17$, sphere $-0.05 \to -0.01$

protocol (plane init):

$\hat{c}[0] = c_{\text{plane}} = +0.037$

$\sigma_{c,0} = 0.5$ (wall family $-0.06 \cdots + 0.64$, two-sided)

$\hat{\bar{a}}_w[0] = D_\perp\Big(\dfrac{1}{\bar{w}[0] - \bar{w}_s}\Big)$

(plane polynomial at the start height: the only value law used)

$P[0] = \text{diag}\big(P_{11}[0],\ P_{22}[0],\ P_{33}[0],\ P_{44}[0],\ \sigma^2_{c,0}\big)$

$Q_{55} = 0$

$P_{44}[0] = \Big(\dfrac{\partial \bar{a}}{\partial \bar{w}_s}\Big)^2 \sigma^2_{\bar{w}_s} + P_{a,\text{floor}}$

(production container; $\sigma_{\bar{w}_s} = 0$ under the protocol)

guard:

$\hat{h}[k] \geq 1$ (the slope is evaluated at $\max(\hat{h}[k],\ 1)$)

$1 - \hat{c}[k] \geq$ gap floor, band $\hat{h} \geq 1.1$

pre-registered pass (wall-test protocol, canon deep, 10 seeds,

$e_{\bar{a}} = \bar{a}_{\text{true}} - \hat{\bar{a}}_w$ absolute):

hold $|e_{\bar{a}}| \leq 0.02$ on plane, cell, sphere; cell descent worst ≤ 0.05;

plane hold within 0.005 of production; $\hat{c}$ honesty 0.7–1.4